TerraPulse: Customer, Persona, Feature and Business Benefit Note

Prepared from project-folder materials Product: TerraPulse mineral intelligence and exploration platform **Developer:** Smart Innovations Ghana (SIG) Ltd **Market focus:** Ghana and West Africa initially, with potential for broader markets

1. Executive summary

TerraPulse is a proposed B2B software and services offering for mineral exploration. It brings satellite, geophysical, geochemical and structural information into a shared geospatial workflow, then helps geologists identify, rank and explain prospective targets. Its commercial promise is to reduce the delay and manual effort between receiving survey data and making a defensible field or drilling decision.

The central business value is **better allocation of exploration capital**: reduce avoidable data-preparation work, help teams compare evidence before mobilizing, and give decision makers clearer support for drilling, financing and concession decisions. The platform supports geologists; it does not replace professional geological judgement, field validation, established 3D modelling tools, or formal resource/reserve reporting.

2. Who is the customer, client, user and buyer?

These terms refer to different roles in the sales and delivery relationship:

- **Customer organization:** The company or institution paying SIG. Likely customers include junior and mid-tier mining/exploration companies, geological consultancies, major mining houses and government geological or mining authorities.
- **Client:** In a consulting engagement, the mining company whose property and data the consultant is assessing. A consulting firm may be the TerraPulse account holder while its mining client is the end beneficiary of the analysis. For direct company sales, customer and client are usually the same organization.

- **Economic buyer:** The person with budget authority, commonly the Exploration Manager/VP, CEO/Technical Director, consultancy principal, or agency/ministry sponsor.
- **Daily user and product champion:** Geologists and technical analysts who upload and interpret data. They influence adoption and provide proof of value to the budget owner.
- **Beneficiaries and approvers:** Board members, investors, joint-venture partners, field crews, regulators and communities may benefit from better decisions or clearer reporting, but are not necessarily software users or purchasers.

Customer segments and commercial fit

Customer segment	Typical buying trigger	Likely commercial offer in project materials
Junior exploration company	Needs to screen concessions and prepare a credible drilling or financing decision with a constrained budget	Team Explorer subscription or seasonal Project Pass
Mid-tier mining company	Needs coordinated analysis across multiple projects, users and commodities	Company Pro subscription; potentially enterprise services
Independent consultant / geological services firm	Wants to deliver more client work with less data setup and reporting effort	Geologist Solo or Company Pro, depending on team and project volume
Major producer	Needs governance, security, model calibration and integration with internal data systems	Enterprise deployment and services
Government / geological survey / regulator	Wants regional mineral intelligence, cadastre visibility, investment materials or environmental oversight	Enterprise/government contract and implementation services

3. Personas and persona-specific value

A. Exploration Manager / VP of Exploration — primary economic buyer

Job: Set exploration priorities, manage geoscience teams and concessions, allocate field and drilling budgets, and defend target choices to executives and joint-venture partners.

Pain points: Geophysical, geochemical and satellite data arrive in inconsistent formats; manual GIS preparation delays decisions; several anomalies compete for limited drilling funds; progress across concessions and teams is difficult to compare.

Most relevant features: Multi-source upload and harmonization; interactive multi-layer map; anomaly detection and ranked targets; evidence matrix showing supporting, conflicting and missing evidence; project dashboards and team permissions; target report generation.

User benefit: Faster comparison of prospects, less repetitive data wrangling, a shared view of project evidence, and a traceable rationale for target selection.

Financial/business benefit: Lower external data-preparation and consultant spend; less risk of committing drilling and mobilization budgets to weak targets; more disciplined allocation across concessions; clearer technical accountability to leadership and partners. These benefits are potential avoided costs, not guaranteed savings or proof that a target will succeed.

Buying case: One avoided unnecessary drill decision or a materially better-ranked campaign could outweigh a software fee, but buyers will require evidence from their own data and field outcomes.

B. Junior Mining CEO / Technical Director — executive sponsor and budget approver

Job: Protect cash runway and shareholder value, approve exploration spend, oversee technical disclosure and financing, and communicate progress credibly to boards and investors.

Pain points: Exploration capital is scarce; dry holes can consume a large share of the budget; lengthy external reports are costly; geological evidence is hard to communicate to non-specialists.

Most relevant features: Ranked target prospectus and board-ready maps/reports; portfolio/concession screening; multi-commodity templates for early property assessment; audit-friendly evidence and methodology; exports to existing technical workflows.

User benefit: A concise, evidence-backed account of why a target or property merits more work, with faster preparation of executive and investor materials.

Financial/business benefit: Better control of exploration spend; reduced time and cost assembling investor or board packages; improved ability to prioritize or relinquish ground before avoidable holding costs; stronger decision support for option, farm-in or acquisition decisions. TerraPulse reports should support—not substitute for—competent-person review and required disclosure standards.

Buying case: Buy when a target decision, financing round or portfolio review is imminent and the cost of delay or poor prioritization is material.

C. Resource / Senior Project Geologist — technical evaluator and core user

Job: Integrate historic and new datasets, interpret geology, generate and vet targets, plan follow-up work, and preserve the reasoning behind technical decisions.

Pain points: Repetitive file cleaning and coordinate conversion; fragmented map and spreadsheet workflows; opaque algorithmic outputs; knowledge lost between office, field and project handovers.

Most relevant features: Multi-modal ingestion; layer toggling and spatial inspection; explainable target scores and confidence; deposit-model templates; geologist feedback/annotation; contextual data query tools where available; export of maps, tables and target coordinates.

User benefit: More time for interpretation and validation, faster review of evidence, and the ability to accept, modify or reject machine-generated suggestions.

Financial/business benefit: Reduced analyst hours and duplicate processing; lower need for ad hoc specialist processing; fewer low-evidence targets advancing to expensive follow-up. The return is strongest when time saved is redeployed to higher-value interpretation rather than treated as automatic headcount savings.

Buying influence: A trusted geologist is often the product champion. Adoption depends on geological transparency, data quality, reproducibility and a useful export path into existing tools such as QGIS or specialist modelling software.

D. Independent Consulting Geologist / Geological Services Firm — customer and channel partner

Job: Deliver target generation, data compilation and technical work for multiple mining clients while protecting margins and reputation.

Pain points: Every client dataset requires setup and cleanup; report preparation consumes billable and non-billable time; desktop software and hardware add overhead; client confidentiality must be protected.

Most relevant features: Separate client workspaces and access controls; reusable processing workflows; standardized target reports; cloud/browser access; shareable exports; project-level audit trail.

User/client benefit: The consultant can turn around analyses faster and provide a consistent, explainable deliverable; the mining client receives a clearer basis for follow-up decisions.

Financial/business benefit: Higher utilization of consultant time, improved project margins, capacity to serve more engagements without equivalent headcount growth, and reduced workstation/software costs where TerraPulse genuinely replaces a paid tool. Client isolation and permissions are critical to this case.

Commercial role: Consultants can be direct subscribers, implementation partners and referral channels into junior and mid-tier companies.

E. Field Exploration Geologist / Ground-Validation Team — daily user and beneficiary

Job: Visit remote concessions, ground-truth anomalies, collect samples, inspect access and constraints, and relay observations to the office team.

Pain points: Poorly transferred coordinates and static maps; limited connectivity; targets may be inaccessible or constrained by terrain, land use or permits; field observations do not always flow back into the interpretation.

Most relevant features: KML/CSV/GeoJSON or GPS-compatible target exports; offline map or cache capability if implemented; access and constraint layers; field annotations and synchronization; clear coordinates and target context.

User benefit: More usable field packages and less manual coordinate transcription; better connection between office targeting and field evidence.

Financial/business benefit: Fewer wasted field days and avoidable mobilization or standby costs; reduced risk of staking, sampling or planning at the wrong location; less rework after field observations are incorporated.

Product caveat: The MVP requirements describe basic map caching, while fully offline-first workflows are listed as out of scope or later-stage in some documents. Do not sell complete offline operation until the shipped capability is confirmed.

F. Government Regulator / Geological Survey Authority — institutional customer

Job: Steward national geological and mineral-rights information, support responsible investment, monitor compliance and environmental impacts, and provide reliable information for policy and licensing decisions.

Pain points: Legacy maps and datasets may be difficult to search or combine; cadastre boundaries and geology may sit in separate systems; regional assessment and investor-facing material can take substantial time; remote monitoring may be limited.

Most relevant features: Regional data integration and prospectivity mapping; cadastre/concession overlays and system integration; searchable national datasets; exportable investment dossiers; satellite change detection and environmental monitoring where validated.

User/public benefit: Faster access to evidence for regional planning and licensing review; improved packaging of public geological information; more consistent spatial oversight.

Financial/business benefit: Reduced external study and data-conversion costs; better use of existing public data; lower administrative effort; improved ability to attract responsible investment and identify compliance issues. Any revenue, fee recovery or environmental savings depend on policy, enforcement and institutional integrations, not software alone.

Buying conditions: Procurement cycles, data ownership, security, residency, interoperability, statutory authority and service-level support will matter more than a self-serve SaaS workflow.

4. Product features mapped to benefits

Feature group	What it does	Main personas	Operational benefit	Financial/business relevance
Data ingestion and harmonization	Accepts supported geophysical grids, geochemistry tables and satellite/geospatial files; aligns them for analysis	Managers, geologists, consultants, agencies	Less manual format conversion and spatial setup	Lower preparation costs and faster time to decision
Unified map and layer analysis	Displays project evidence together with controls to inspect and compare layers	Geologists, managers, field teams	Shared spatial context; fewer disconnected tools and static exports	Reduced analyst effort and rework; may reduce some software overhead
Prospectivity and anomaly ranking	Flags and prioritizes areas using available data and configured models	Geologists, managers, CEOs, agencies	Focuses review and follow-up on the strongest combined evidence	Better allocation of scarce sampling, survey and drilling budgets

Feature group	What it does	Main personas	Operational benefit	Financial/business relevance
Explainability and evidence matrix	Shows what evidence supports or weakens a target and what is missing	Geologists, technical directors, managers	Enables expert review rather than blind acceptance	Reduces the risk of advancing poorly supported targets; improves defensibility
Reports and target exports	Creates maps, target summaries and coordinate files for decisions or field workflows	CEOs, managers, consultants, field crews	Reduces manual report assembly and coordinate transfer	Lower delivery effort; faster internal approvals and field mobilization
Team and project workspaces	Organizes projects, roles, access and client data	Managers, consultants, companies, agencies	More consistent collaboration and handover	Supports team subscriptions and repeatable consulting delivery
Portfolio / multi-commodity tools	Compares projects, concessions or mineral-system evidence	CEOs, technical directors, portfolio managers	Faster screening and prioritization	Can reduce time spent carrying low-priority ground or pursuing weak opportunities
Offline and field capabilities	Makes selected maps/data usable in remote areas and captures observations	Field teams, exploration managers	Less dependence on connectivity and office GIS	Potentially fewer wasted field hours; implementation status must be verified
Enterprise integrations and deployment	Connects to institutional data systems and offers governed deployment	Major companies, regulators, surveys	Fits internal security and data processes	Enables larger contracts; carries implementation and support costs

5. Financial logic: how value is created

TerraPulse's financial value is indirect and risk-based. The platform does not create revenue simply by producing a map. Value is realized when its analysis changes a decision or reduces work:

1. **Preparation-cost reduction:** Fewer paid hours or external invoices for data cleaning, projection, gridding and map assembly.
2. **Time-to-decision reduction:** Earlier target or portfolio decisions; less waiting for data to be made comparable.
3. **Capital-at-risk reduction:** Better evidence review before costly surveys, access work, trenching, drilling or option payments.
4. **Capacity and margin improvement:** Consultants and internal teams complete more useful work with the same staff time.
5. **Governance and financing support:** More traceable assumptions and clearer technical materials for boards, partners, investors or agencies.

Illustrative ROI formula

Net value for a period = verified avoided external processing cost + verified labor capacity released + expected cost of avoidable work prevented – TerraPulse fees – onboarding/integration costs.

For drilling-risk decisions, use an expected-value calculation rather than claiming every avoided hole as a saving:

Expected drilling value = (change in probability of an avoidable decision × cost of that decision) – incremental analysis and validation cost.

An illustrative project-folder example uses a 250 m hole at $250/m$(62,500), 100 analyst hours valued at $7,500and a7$,990 annual Team Explorer fee, producing a notional $62,010 net benefit. This is an example, not a measured customer result: it assumes the platform actually prevents that hole, that the hours are truly saved, and excludes onboarding, field validation, and other costs.

6. Pricing references in the folder

The later commercialization model proposes:

Offer	Proposed price	Intended segment
Geologist Solo	$199/monthor$ 1,990/year	Independent consultants and solo professionals
Team Explorer	$799/monthor$ 7,990/year	Junior exploration companies
Company Pro	$2,200/monthor$ 22,000/year	Mid-tier companies and larger consultancies
Campaign Project Pass	$2,500$–4,500 for 90–120 days	Seasonal explorers
Enterprise / Government	$50,000$–150,000+ per year	Major producers and public institutions
Data modernization	$5,000$–25,000 one-off	Organizations with legacy data

An earlier Lean Canvas gives different indicative tiers ($150,$600 and $1,800per month)and a consultant rate of$400 per project. Treat pricing as a hypothesis to validate, not a finalized price list. Confirm which plan, feature entitlement, taxes, support and implementation terms are current before external use.

7. Which persona to prioritize

The strongest initial account is a **junior exploration company** with an active concession, real data ready for analysis, and a near-term target or drilling decision. Start with the Exploration Manager as economic buyer and a Senior/Project Geologist as the daily champion. The CEO or Technical Director approves the spend when the output supports a material capital allocation or financing decision.

Consultants are a useful second route: they can adopt the product directly and expose it to multiple mining clients. Field geologists should be included in workflow validation because target utility depends on successful handoff and ground truth. Government and major-enterprise opportunities may have high contract values but are likely to need longer procurement, integration and security work.

8. Claims and validation discipline

Project documents quote savings such as weeks reduced to minutes, six-figure drilling risk avoided, and specific annual savings. These figures are **proposed value hypotheses** unless backed by measured pilot data. Before presenting them as customer outcomes, measure:

- Baseline time and external cost for data preparation, target review and report production.
- User time and total elapsed time using TerraPulse, including correction and validation.
- Whether ranked targets change field or drilling decisions, and why.
- False-positive and missed-target rates against independent geological review and ground truth.
- Total cost of ownership, including subscriptions, onboarding, compute/data charges and support.
- Renewal willingness and budget owner approval at the proposed price.

Avoid saying the product guarantees discoveries, eliminates dry holes, produces a mineral resource/reserve estimate, or creates compliant JORC/NI 43-101 disclosure by itself. Describe it as decision support that organizes evidence and helps prioritize follow-up.

9. Source documents used

- `03 STAGE 3 EARLY TRACTION/Early Access & Commercialization/VALUE_PROPOSITION_PER_PERSONA_AND_FEATURES.md`
- `03 STAGE 3 EARLY TRACTION/Early Access & Commercialization/VALUE_PROPOSITION_AND_PRICING_MODEL.md`
- `01 STAGE 1 IDEA VALIDATION/02_User_Personas.md`
- `01 STAGE 1 IDEA VALIDATION/03_Lean_Canvas.md`
- `01 STAGE 1 IDEA VALIDATION/05_PRD_MVP.md`